Artifical Intelligence Basics Course Summary

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1 Key Terms

- **Minimax Search**: A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally.
- **Alpha-Beta Pruning**: A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision.
- **Turing Test**: An operational test for intelligent behavior, the Imitation Game, where a human interrogator tries to distinguish between a human and a machine based on their text-based responses.
- **Game Tree**: A tree-like structure representing all possible game states and moves, starting from the initial state and branching out to all possible outcomes.
- **Perceptron**: A fundamental unit in neural networks, consisting of weighted inputs, a summation node, and an output, used for binary classification.
- **Reinforcement Learning**: A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.
- **Matrix**: A rectangular array of numbers, symbols, or expressions arranged in rows and columns.
- **Linear Transform**: A function that maps vectors from one vector space to another, satisfying $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all vectors u, v and scalars c.
- **Matrix Multiplication**: An operation that combines two matrices to produce a third matrix, where the element at row i and column j of the resulting matrix is the dot product of row i of the first matrix and column j of the second matrix. The inner dimensions must match.
- **Matrix Inverse**: For a square matrix A, its inverse A^{-1} is a matrix such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.
- **Matrix Transposition**: An operation that swaps the rows and columns of a matrix. The transpose of matrix A is denoted as A^T.
- **Derivative**: A measure of how a function changes in response to a small change in its input. It represents the instantaneous rate of change.
- **Partial Derivative**: A derivative of a function of multiple variables with respect to one of those variables, holding the others constant.
- **Sigmoid Function**: A mathematical function that maps any input value to an output value between 0 and 1. Often used as an activation function in neural networks.
- **Inductive Reasoning**: The process of deriving general principles from specific observations.
- **Linear Regression**: A machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.

- **Model Evaluation**: The process of assessing the performance of a machine learning model.
- **Overfitting**: A phenomenon in machine learning where a model learns the training data too well, resulting in poor generalization to unseen data.
- **Representative Data**: Data that accurately reflects the characteristics of the problem domain being addressed.
- **Binary Classification**: A machine learning task where the goal is to classify data into one of two categories.
- **Decision Boundary**: A line or surface that separates different classes of data points in a classification problem.
- **0-1 Loss**: A loss function in binary classification that assigns a loss of 1 if the prediction is incorrect and 0 if it is correct.
- **Perceptron Loss**: A loss function used in the perceptron algorithm, defined as max0, -z, where z = y(w^Tx).
- **Hinge Loss**: A loss function used in support vector machines, defined as max0, 1 - z, where z = y(w^Tx).
- **Logistic Loss**: A loss function used in logistic regression, defined as log(1 + e^{-z}), where z = y(w^Tx).
- **Gradient Descent**: An iterative optimization algorithm used to find the minimum of a function by following the direction of the negative gradient.
- **Stochastic Gradient Descent**: A variant of gradient descent that updates the model parameters using a randomly selected subset of the data.
- **Logistic Regression**: A machine learning algorithm used for binary classification that models the probability of a data point belonging to a particular class using the logistic function.

2 Problem Types

- **Understanding the Nature of Intelligence**: This problem explores the definition and characteristics of intelligence, both human and artificial. It examines various perspectives on what constitutes intelligence and how it can be measured or simulated.
- **Defining Artificial Intelligence**: This problem involves formulating a precise definition of artificial intelligence, considering different perspectives and historical contexts. The challenge lies in capturing the essence of AI while accounting for its evolving nature and diverse applications.
- **Passing the Turing Test**: This problem focuses on the implications of passing the Turing Test, a benchmark for machine intelligence. It explores whether passing the test truly indicates machine intelligence or merely the ability to mimic human behavior.
- **Overcoming the Limitations of Search**: This problem addresses the computational challenges associated with exhaustive search in complex games like chess and Go. It explores the need for alternative approaches, such as machine learning, to overcome the limitations of brute-force search.
- **Addressing the Challenges of Computer Vision**: This problem investigates the difficulties in developing computer vision systems that can reliably interpret and understand images. It highlights the gap between initial expectations and the ongoing challenges in this field.
- **Mitigating AI Safety Risks**: This problem focuses on the potential risks associated with AI systems, such as adversarial attacks and biases. It explores methods for improving the safety and reliability of AI systems.

- **Addressing Ethical Concerns in AI**: This problem examines the ethical implications of AI systems, particularly concerning bias and fairness. It explores the need for responsible AI development and deployment to mitigate potential harm.
- **Understanding the Differences Between Game Types**: This problem explores the differences between various game types, such as chess and soccer, in terms of their complexity, dynamics, and the challenges they pose for AI algorithms.
- **Bridging the Gap Between Different AI Domains**: This problem investigates the similarities and differences between seemingly disparate AI domains, such as game playing and speech synthesis, to identify common principles and approaches.
- **Developing Robust and Generalizable Algorithms**: This problem focuses on the design of algorithms that can adapt to new situations and generalize their knowledge to unseen scenarios. It explores the challenges in creating algorithms that are not only effective but also robust and adaptable.
- **Understanding the AI Grand Challenges**: This problem explores the reasons behind selecting specific problems, such as autonomous driving, as major challenges in the field of AI. It examines the factors that make these problems particularly significant and difficult to solve.
- **Designing Effective Game-Playing Systems**: This problem focuses on the design of algorithms for playing games, considering various components such as state representation, action selection, and opponent modeling.
- **Understanding the Architecture of Advanced AI Systems**: This problem explores the architecture and design of complex AI systems, such as AlphaGo, which combine deep learning and search techniques to achieve superhuman performance.
- **Representing Linear Transformations with Matrices**: Matrices are used to represent linear transformations, which map vectors from one space to another while preserving linear combinations. This is fundamental to many AI algorithms.
- **Matrix Arithmetic**: The slide covers basic matrix operations such as addition, subtraction, scalar multiplication, and multiplication, which are essential for manipulating data in AI applications.
- **Solving Systems of Linear Equations**: Systems of linear equations can be represented and solved using matrices and their inverses. This is crucial for various AI tasks involving optimization and data analysis.
- **Matrix Transposition**: Matrix transposition is a fundamental operation that involves swapping rows and columns of a matrix. This is used for data manipulation and transformation in AI.
- **Understanding Perceptrons**: The perceptron, a fundamental building block of neural networks, is introduced. Its structure and function are explained using a diagram.
- **Parallelizing Matrix Multiplication**: The slide demonstrates how matrix multiplication can be parallelized using domain decomposition, improving computational efficiency in AI applications.
- **Calculus in AI**: The slide briefly mentions the importance of calculus, specifically derivatives, partial derivatives, and finding maxima/minima/gradients, in AI algorithms.
- **Calculating the Derivative of the Sigmoid Function**: The slide shows the step-by-step derivation of the derivative of the sigmoid function using calculus rules. This is important for training neural networks.
- **Linear Regression**: Finding the line that best fits a set of data points, minimizing the sum of squared distances between the points and the line.
- **Multiple Linear Regression**: Extending linear regression to include multiple independent variables to predict a dependent variable.

- **Curve Fitting**: Fitting a curve to a set of data points, potentially using polynomial functions of varying degrees.
- **Overfitting**: The phenomenon where a model fits the training data too closely, resulting in poor generalization to unseen data.
- **Binary Classification**: Classifying data points into one of two categories using a learned function.
- **Choosing a Loss Function**: Selecting an appropriate loss function (e.g., 0-1 loss, perceptron loss, hinge loss, logistic loss) to measure the error of a classification model.
- **Minimizing Loss using Gradient Descent**: Iteratively updating model parameters to minimize the chosen loss function using gradient descent or stochastic gradient descent.
- **Comparing Perceptron and Logistic Regression**: Contrasting the Perceptron algorithm and logistic regression, highlighting their update rules and differences in approach to binary classification.

3 Algorithm Solutions

- **Minimax Search**: A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning**: An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.
- **Matrix Addition**: $A + B = C$, where $c_{ij} = a_{ij} + b_{ij}$
- **Matrix Scalar Multiplication**: $k \cdot A = B$, where $b_{ij} = k \cdot a_{ij}$
- **Matrix Multiplication**: $C_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$
- **Matrix Transposition**: Swapping rows and columns of a matrix A to obtain A^T.
- **Solving Linear Equations with Invertible Matrices**: $Ax = bx = A^{-1}b$
- **Derivative of Sigmoid Function**: $\frac{d}{dx}\sigma(x) = \frac{e^{-x}}{(1+e^{-x})^2}$
- **Perceptron Algorithm**: Iteratively update the weight vector w by adding $y_n x_n$ if the data point x_n is misclassified, where y_n is the true label. Repeat until convergence.